Project TIWALA: A User-Centered Framework for Farmer Empowerment in Isabela, Philippines

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***Abstract*— Farmers in Isabela, Philippines, face significant threats to their livelihoods from adverse weather conditions and volatile market prices, compounded by a critical communication gap where vital information is often too technical to be useful. This paper introduces Project TIWALA, a user-centered framework for farmer empowerment that addresses this challenge. The framework's design is grounded in a detailed User-Centered Design (UCD) methodology and guided by a novel conceptual model that extends the Technology Acceptance Model (TAM) with the constructs of Trust and Social Influence. Crucially, the entire system is architected upon a "Safety by Design" (SbD) philosophy to create a secure, trusted digital ecosystem for a vulnerable user base. By leveraging cost-effective AI and HCI principles, Project TIWALA offers a foundational model for safely and effectively democratizing agricultural intelligence.**

Keywords*— *Agricultural Technology, Agriculture Intelligence, Human-Computer Interaction, Technology Acceptance Model, Safety by Design, User-Centered Design, Mobile Applications, Rural Development, Livelihood Support and Protection, Isabela Province, Philippines

I. Introduction

The rice industry in the Philippines, which contributes around 385.59 billion pesos to the country's overall economy, serves as a fundamental pillar for national food security. This sector is sustained by the efforts of approximately 2.87 million farmers who play a crucial role in its operation. Although the industry holds considerable economic and social value, it operates with extremely slim profit margins. On average, the national net return amounts to only 36,211 Philippine pesos per hectare for each growing season.

In Isabela province, the "Rice Bowl of the North," this challenge is even more acute. Despite producing 784,843 metric tons of palay in the first semester of 2025, the province's 113,038 rice farmers face a stark economic reality, with a net income of only 15,982 PHP per cropping season, and at this day and age, this is a stretch, especially for a family of five. This is why our project is a strategic intervention to enhance profitability and aims to support the livelihoods of the farming community.

This financial instability is exacerbated by the growing unpredictability of climate conditions and the complexities of "social weather" the fluctuating market prices, government initiatives, and technological recommendations that farmers must manage. The core problem, and the central research gap this paper addresses, lies in the "last-mile communication" of critical information. While government agencies like the Department of Agriculture (DA), DOST, and BSWM generate vast amounts of valuable data, this information often fails to reach the farmer in a timely, understandable, and trustworthy format. This leads to a farming culture that responds reactively, with choices grounded in traditional methods that are becoming less dependable amid contemporary challenges.

The research initiative, TIWALA (Temporal and Integrated Weather Analytics for Local Agriculture), suggests a mobile-first decision support system (DSS) as its approach. The project is driven by three fundamental HCI questions:

1. How can UI/UX design, specifically an empathetic and affective interface, effectively translate complex, technical data into simple, actionable advice that farmers will trust and adopt?

2. How can this information be communicated to farmers with limited resources and varying levels of digital literacy?

3. How can we verify that the information is received, understood, and perceived as valuable by the end-user?

This paper will detail the use of the User-Centered Design (UCD) methodology employed in developing the TIWALA framework. We will discuss the related work in technology adoption and existing government systems, present the system's architecture and key features derived from direct user feedback, its architecture built upon a "Safety by Design" philosophy (SbD) [6], ensuring that the digital well-being of its users-particularly an older demographic less familiar with the technology--is prioritized, and finish by discussing the wider significance of creating designs that build trust to encourage proactive resilience in agriculture.

II. Related Work

The creation of the TIWALA framework is guided by three main research areas: technology adoption theories in HCI, the current governmental and legal structures related to rice farming, and the particular technological environment in the province of Isabela, which is our focus.

A. Technology Adoption in HCI

For an ICT system to be effective, it must first be adopted. The Technology Acceptance Model (TAM) offers a basic structure for understanding the elements that affect a user's choice to embrace a new technology.

TAM suggests that user acceptance mainly depends on how useful and easy to use the technology is perceived to be.

Given that the average age of Filipino farmers is 58, with 74% belonging to the Baby Boomer and Generation X groups, it is essential that tools are easy to use. Moreover, extending

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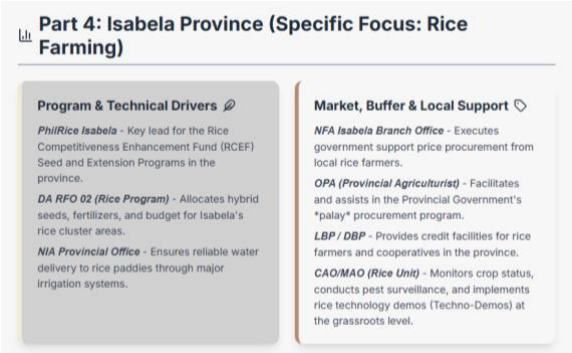
from TAM, later studies have highlighted the importance of social influence and trust, especially within communities dependent on peer-to-peer knowledge exchange. Consequently, any effective intervention needs to be not only straightforward and beneficial but also seen as trustworthy and consistent with community standards.

B. Governmental and Legal Framework for Rice Farming

The Philippine rice sector is governed by a multi-layered network of government agencies operating under the Republic Act No. 11203, or the Rice Tariffication Law. This law established the Rice Competitiveness Enhancement Fund (RCEF), which directs funding towards improving the productivity of rice farmers. The ecosystem is led by the Department of Agriculture (DA) and its National Rice Program. Key specialized support functions are carried out by attached agencies like PhilRice (R&D), PhilMech (Mechanization), and the ATI (Training). Other critical national bodies like the NIA, the NFA, and the LBP manage irrigation, buffer stocking, and credit, respectively.

Figure 1: A visual representation of the multi-agency framework for government support in the Philippine rice sector. While comprehensive, this illustrates the fragmented information landscape that contributes to the 'last-mile' communication gap.



Although these agencies offer extensive support systems, they frequently lead to information silos, highlighting the importance of having a unified communication platform.

C. *Current Initiatives and the Persisting Information Gap*

Recent developments within Isabela demonstrate a significant push towards agri-innovation. Massive investments are being made in infrastructure, such as the Philippines' largest solar-powered pump irrigation project. This "hardware" support is complemented by aid distributed by various agencies: engine sets from the DA, toolkits from TESDA, and cash assistance from DSWD. However, these initiatives, while vital, are often reactive and fragmented. The devastating P9.89 billion in agricultural losses from the 2024 El Niño event highlight the urgent need for proactive strategies. This is the precise gap Project TIWALA addresses. It serves as the informational and decision-making layer that synthesizes advisories and translates complex climate data, allowing farmers to make the most efficient use of these dispersed resources.

D. *Technology Landscape in Isabela*

The technological readiness for such a system is already in place. Data shows that 96% of Isabela's farmers own a smartphone. Furthermore, the community has strongly embraced mechanical innovation, with adoption rates for hand tractors (97.4%) and combine harvesters (87.9%) being exceptionally high. This indicates a willingness to invest in technologies that provide clear benefits. Nevertheless, the uptake of more sophisticated, data-driven methods such as controlled irrigation (39.7%) remains limited. This gap presents a clear opportunity for a solution like TIWALA, which aims to augment the intellectual aspects of farming rather than substitute mechanical labor.

III. Methodology

The design and conceptualization of Project TIWALA are rooted in a User-Centered Design (UCD) process. This approach, which heavily employs Interaction Design principles, ensures that the resulting socio-technical framework is not only technologically sound but also usable, effective, and empathetic to the needs of its target users: the farmers of Isabela. The methodology was structured into three distinct phases as outlined below.

A. *Phase 1: Problem and User Space Exploration*

The initial phase focused on foundational research to deeply understand the context and the users.

1. Problem Space Exploration - Our initial investigation confirmed a dual crisis facing farmers in Isabela. The region is subjected to approximately 20 typhoons annually, creating a critical window of vulnerability. This climate-related threat is compounded by a severe economic crisis. User interviews and survey data consistently revealed that the inflated cost of inputs, low market price of palay, and a perceived lack of government support are primary sources of distress and financial instability for the farming community.

2. User Space Exploration - To understand the farmers' perspectives, we employed a mix of qualitative and quantitative methods, including direct observations and a survey of 49 farmers from the community.

The exploration revealed crucial insights into their information-seeking behaviors and technology usage. Critically, 44.9% of farmers rely on "Fellow farmers" for information, highlighting the importance of community trust. Furthermore, the survey showed that 73.47% of farmers own a smartphone, indicating a strong existing digital footprint. To better illustrate these foundational findings, the following visualizations are presented:

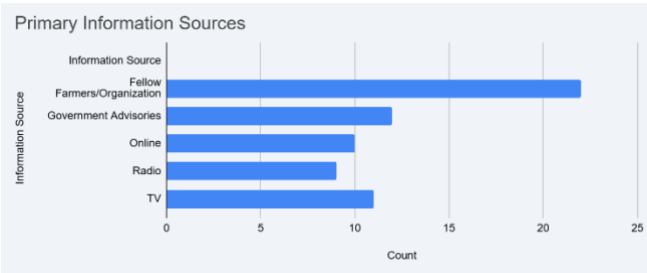


Figure 2: Bar Chart of Primary Information Sources. This chart visually emphasizes the dominance of "Fellow Farmers/Organization" compared to other sources like "Government Advisories," "TV," and "Online," reinforcing the need for social and trust-based features.

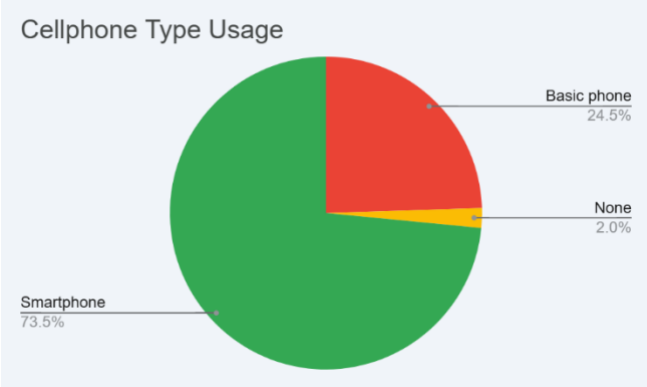


Figure 3: Pie Chart of Cellphone Type Usage. This chart illustrates the breakdown between "Smartphone" users (73.5%) and "Basic Phone" users, visually justifying the necessity of a multi-channel approach that includes both a low-data app and a robust SMS service.

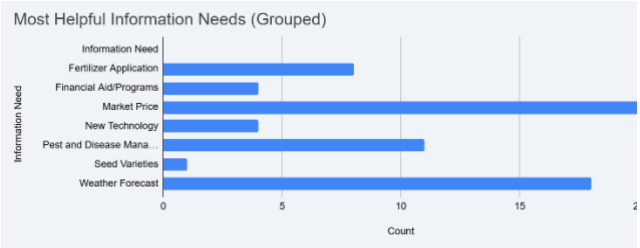


Figure 4: Bar Chart of Most Helpful Information. This chart ranks the information needs of farmers, clearly showing "Weather forecast" and "Market price of rice" as the top two priorities, justifying their prominence in the app's User Experience (UX) design choices.

B. Phase 2: Prototyping and Design

The insights gathered in Phase 1 directly informed the design of the TIWALA application. The prototyping process was iterative and user-focused, beginning with low-fidelity wireframe sketches and progressing to high-fidelity, interactive prototypes that were tested with 30 farmers. This entire design process was guided by a SbD philosophy [6], ensuring that features were not only useful and usable but also explicitly protected farmers from potential misinformation and digital harms.

- Perceived Usefulness (PU) - The survey overwhelmingly identified Weather Forecasts and Market Price of Rice as the two most important pieces of information. Therefore, these features were made the central components of the application's home screen.
- Perceived Ease of Use (PEOU) - Recognizing that farmers prefer receiving timely information via text (SMS) or mobile apps, the UI/UX design prioritized simplicity with large, clear icons and minimal text.

C. Phase 3: System Evaluation

To measure the potential success and impact of Project TIWALA, a set of clear success metrics was defined, including Usability, Adoption, and Empowerment, with a key usability benchmark being the ability for a farmer to get a clear recommendation in under 60 seconds. Initial tests confirmed this was achievable.

TABLE I. USABILITY TEST RESULTS FOR CORE APP FEATURES

Feature	**(Smartphones)**		
	Basic	***Standard***	***High-End***
Weather Forecast Check	35.18s	28.45s	22.81s
Market Price Inquiry	38.74s	31.91s	25.01s
Govt. Bulletin Access	43.10s	36.62s	31.54s

Note: We conducted 30 tests, and the duration represents the average time for the feature when operating under the weakest internet signal condition.

D. AI and NLP Implementation Strategy

The TIWALA framework will leverage AI not as a complex, user-facing technology, but as a powerful backend engine to simplify and personalize the user experience.

- NLP for Information Simplification: Government advisories and technical bulletins are often published in dense, jargon-filled documents. An NLP summarization model will be trained to ingest these documents and automatically generate concise, easy-to-understand alerts in the local language (e.g., Ilocano, Tagalog). For example, a 10-page report on fertilizer subsidies could be distilled into a single SMS: *"Mag-apply para sa libreng abono bago mag-Oct 30. Dalhin ang inyong ID sa munisipyo. - DA"* (Apply for free fertilizer before Oct 30. Bring your ID to the municipal hall. - DA). This directly addresses the user-reported problem of information being "too technical."
- NLP for Sentiment Analysis *("Social Weather")*: The "Bayanihan" feature will serve as a source of real-time ground truth. An NLP sentiment analysis model will process the anonymized text from community discussions to gauge farmer sentiment (positive, negative, neutral) and identify recurring keywords (e.g., "*peste*," "*presyo*," "*patubig*"). This provides a "social weather" report for agricultural agencies, allowing them to respond proactively to widespread issues.
- Machine Learning for Personalization: A simple ML model will learn user preferences based on their interactions. For instance, a user who frequently taps on "Market Price" alerts will have that information prioritized in their main feed. Similarly, users in a specific municipality known to have irrigation issues could automatically receive more frequent water schedule advisories. This ensures that the most relevant information is always surfaced first, increasing the app's Perceived Usefulness.

IV. CONCEPTUAL FRAMEWORK: THE TIWALA TECHNOLOGY ACCEPTANCE MODEL

To ground the design and evaluation of Project TIWALA in established theory, we adopt and extend TAM [4]. For the specific context of Filipino farmers, we extend this model by introducing Trust and Social Influence as critical moderators and explicitly acknowledge Contextual Constraints.

A. Mapping TAM Elements to the TIWALA Context

1) Perceived Usefulness (PU) - High PU is associated with features providing timely market prices, weather alerts, and subsidy information, as confirmed by our survey.

2) Perceived Ease of Use (PEOU) - Prioritizes an SMS-first approach, simple navigation, and local language to address concerns about usability for older farmers.

3) Trust - Acknowledging the target demographic's average age of 55 and varying levels of technological familiarity, our framework elevates Trust beyond a simple construct to a core design philosophy, adopting a SbD [6] approach. This proactive strategy focuses on safeguarding users from potential digital harms like misinformation and financial scams. For TIWALA, this means engineering a protected ecosystem through:

a) Data Provenance - Clearly displaying the source and validation status of all advisories (e.g., "DA-PhilRice Verified")

b) Verified Community - Restricting the "Bayanihan" network to authenticated farmers and experts to prevent malicious actors and foster a safe space for knowledge sharing.

c) Simplified Choices - Designing user flows that guide farmers toward clear, vetted actions, reducing the cognitive load and potential for costly mistakes. This safety-centric approach is fundamental to building lasting trust with a user base that values proven results and official credibility.

4) Social Influence - With 44.9% of farmers relying on peers, the app enables easy sharing of advisories to farmer groups to leverage existing social networks.

5) Contextual Constraints (External Variables) - *The design addresses barriers like device type and intermittent connectivity through multi-channel delivery (SMS, low-data app) and offline caching.*

B. Farmer Personas and TAM Interpretation

To illustrate the model in practice, we developed three key personas:

1. Persona A – "The Elder Farmer" (65 y/o, basic phone): Addressed with SMS-first alerts and endorsements from trusted local leaders to increase PEOU and social proof.

2. Persona B – "The Mid-Age Smartphone User" (45 y/o, smartphone): Targeted with a clear app dashboard and provenance badges to increase PU and Trust.

3. Persona C – "The Young Tech-Adopter" (30 y/o, smartphone): Engaged with push notifications and automation to maximize both PU and PEOU.

V. PILOT EVALUATION PLAN AND EXPECTED RESULTS

To empirically validate the TIWALA-TAM framework, a structured 8-week pilot evaluation is designed for a sample of 200 farmers in Isabela, stratified by phone type (60% smartphone, 40% basic phone). A mixed-methods approach will be employed, including baseline/endline surveys, real-time behavioral logging, and qualitative interviews.

A. Key Performance Indicators (KPIs) and Targets

To empirically validate the TIWALA-TAM framework, a structured 8-week pilot evaluation is designed for a sample of 200 farmers in Isabela, stratified by phone type (60% smartphone, 40% basic phone). A mixed-methods approach will be employed, including baseline/endline surveys, real-time behavioral logging, and qualitative interviews.

KPI	Target
Reliability	⩾ 95% delivery rate of critical alerts.
Engagement	⩾ 75% read-confirmation rate for critical alerts.
Usefulness	⩾ 40% of users report a changed farming decision.
Ease of Use	⩾ 70% (App) / 45% (SMS) of users completing a guided action without help.
Trust Growth	+0.6 mean increase on a 1–5 trust scale.
Social Uptake	⩾ 30% of users share at least one message.
Retention	⩾ 50% Weekly Active Users (WAU).

B. Key Results Area (KRA) and Expected Results

This section outlines the anticipated results from the 8-week pilot, with hypotheses grounded in our baseline survey data.

KRA	Expected Outcome
Engagement and Reliability	High channel reliability (≥ 95%) is expected due to 100% cellphone ownership. High engagement (≥ 75% read confirmation) is anticipated based on existing habits of sharing text information (38.8%).
Perceived Usefulness (PU)	We hypothesize that the ≥ 40% target for decision changes will be met, as the app's core features directly address the top two user needs: Weather Forecasts (28.6%) and Market Prices (22.5%).
Perceived Ease of Use (PEOU)	We anticipate meeting usability targets (≥ 70% App / 45% SMS) due to high smartphone penetration (73.5%) combined with a simple, user-centric design catering to the average user age of 55.
Trust and Social Influence	We expect to meet the +0.6-trust growth target by building on the high initial trust in government sources (4.47/5) and exceed the 30% share rate target by leveraging the strong reliance on peer networks (44.9%).
Adoption and Retention	We expect to meet the ≥ 50% WAU target, driven by the high willingness to adopt new technology (87.8%) to solve pressing challenges like low produce prices.

VI. DISCUSSION

The expected success of the Project TIWALA pilot carries significant implications for agricultural extension, technology design, and Isabela farmer empowerment.

A. Bridging the Communication Gap

The framework is poised to effectively bridge the communication gap by addressing the key farmer-cited problems of information being "too technical" (via high PEOU) and "unsure if true" (via trust-building features).

B. Validating the Socio-Technical Design Approach

The expected success confirms our extended TAM, confirming that for this context, Trust and Social Influence are critical drivers of adoption. The results would also underscore the necessity of a SbD approach when introducing technology to vulnerable user groups, proving that a protected digital environment is a prerequisite for sustained engagement.

C. Translating Theory into Practice: Key System Features

The app's design directly embodies the framework. Features like the "Pagtitiwala" (Trust) section provide data provenance, while the "Bayanihan" (Communal Unity) feature creates a professional network—a "LinkedIn for farmers." This network's restriction to verified stakeholders is a direct manifestation of our SbD philosophy [6], creating a productive environment free from the misinformation common on open social platforms. The non-profit, government-sponsored model further reinforces trust by ensuring the platform remains unbiased and pro-farmer.

D. Implications for Farmer Empowerment

Ultimately, the significance of Project TIWALA lies in its potential to empower farmers. The expected high rate of decision-changing actions (≥ 40%) translates into tangible benefits, moving farmers from a position of reactive uncertainty to proactive, data-informed decision-making within a safe and trusted digital ecosystem.

E. Limitations and Future Work

The primary limitation is that these results are expected pending the execution of the 8-week pilot. Future work will involve a longitudinal, year-long study to measure impact across harvest cycles and explore the framework's scalability to other regions and crop types.

VII. CONCLUSION

The farmers of Isabela, as producers in one of the country's key agricultural provinces, face a significant information gap that hinders their ability to make timely, data-informed decisions, thereby threatening their livelihoods. This paper introduced Project TIWALA, a conceptual socio-technical framework designed to bridge this gap by democratizing agricultural intelligence. The primary contribution is a holistic framework that presents a detailed UCD methodology, proposes an extended TAM tailored to a rural Filipino context, and details the design of a non-profit application built on a SbD philosophy. By translating complex data into simple, actionable insights, Project TIWALA offers a robust, empathetic, and safe model for technology adoption—a foundational step toward empowering farming communities.

As this paper presents a conceptual framework, its primary limitations must be acknowledged. The proposed pilot study's geographic focus on Isabela and its relatively short 8-week duration may limit the generalizability of the findings to other regions or across different cropping seasons. Furthermore, the evaluation methodology itself is shaped by the practical constraints of the research. Due to limited resources, the plan relies heavily on surveys and behavioral logging. While effective, these methods could not be supplemented by more resource-intensive approaches that could yield deeper insights. For instance, long-term ethnographic studies would provide a richer understanding of cultural integration, and a full-scale randomized controlled trial (RCT) would be required for a more definitive measure of the system's direct economic impact. Consequently, the current plan offers a practical and targeted strategy for initial validation, prioritizing effectiveness over a comprehensive assessment.

Looking forward, we offer several recommendations. For policymakers, we recommend investing in "last-mile" communication infrastructure and mandating a SbD approach for government-supported agricultural technology. For technology developers, we recommend prioritizing socio-technical design and building resilient, multi-channel systems (SMS, low-data app, voice) to ensure inclusivity. Future research should focus on a longitudinal, year-long study to measure the framework's impact across multiple harvest cycles and explore its scalability to other provinces and crop types. By implementing these recommendations, stakeholders can work collaboratively to create a more resilient, informed, and empowered agricultural sector.

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All AI-generated outputs were thoroughly reviewed and significantly edited by the authors, who maintain full responsibility for the paper's content and arguments.

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APPENDIX A: RESEARCH INSTRUMENTS AND PROTOTYPES

This appendix contains the survey questionnaire distributed to farmers and a selection of high-fidelity prototype screens for the TIWALA application.

A.1 Original Survey Instrument

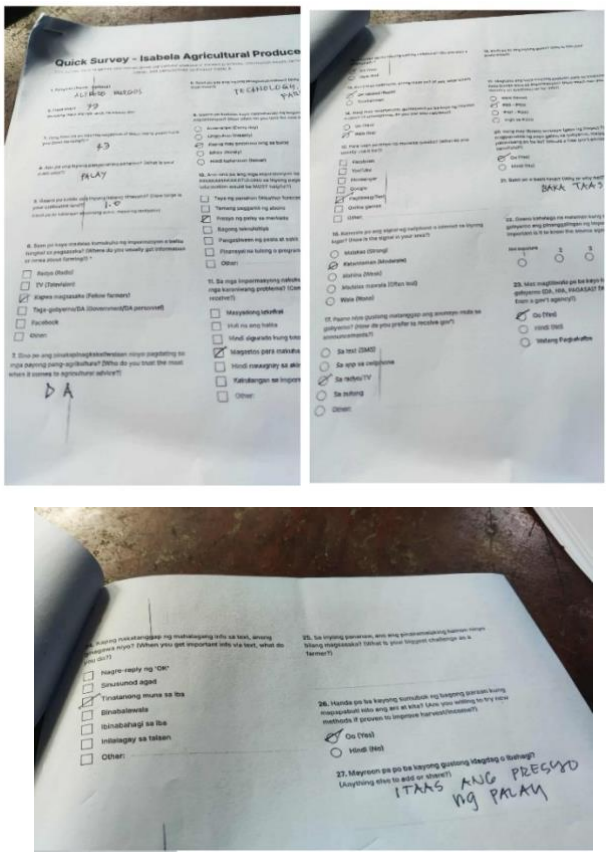

Figure 5: The original survey form distributed to farmers.

A.2. High-Fidelity Prototype Screens

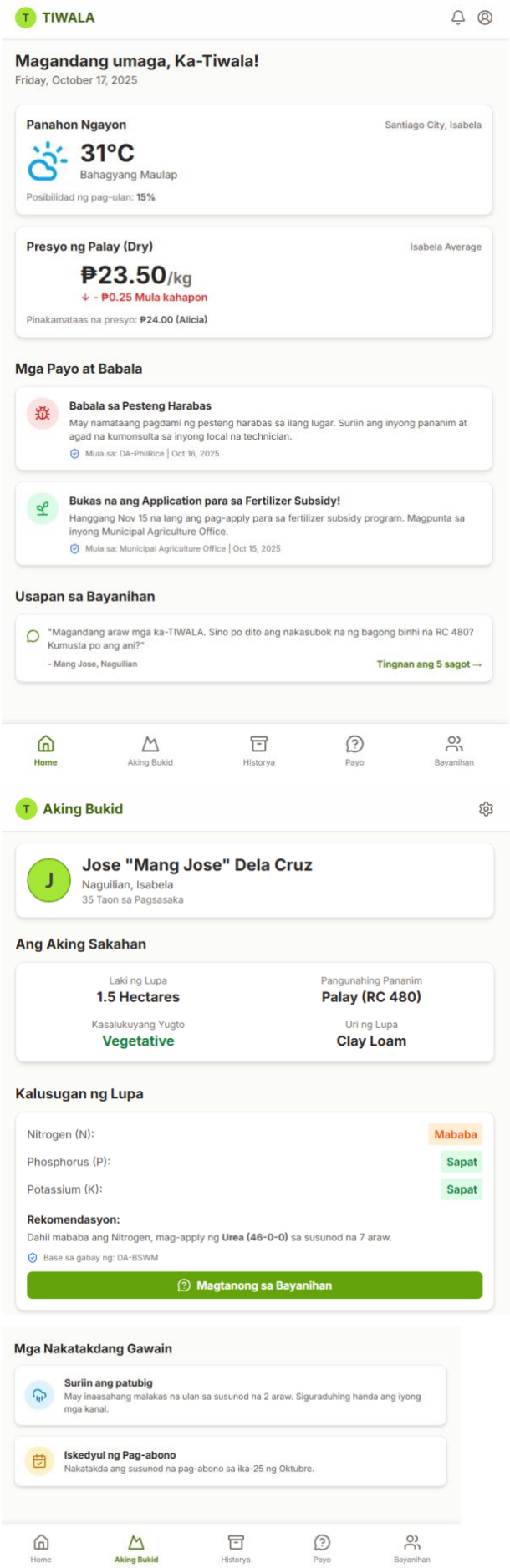

Figure 6: The Main Dashboard screen. This design prioritizes Perceived Usefulness (PU) by displaying the most requested information—weather and market prices—at the forefront. The clean layout with large icons addresses Perceived Ease of Use (PEOU).

Figure 7: The Historical Data ("Historya") screen. This screen builds Trust by providing transparent access to past advisories and seasonal outcomes, allowing farmers to learn from historical trends and verify the system's track record over time.

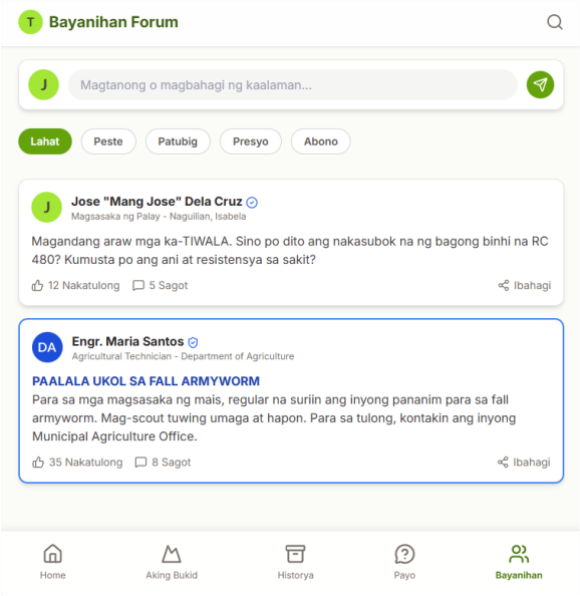

Figure 8: The Community Network ("Bayanihan") screen. This feature is designed to leverage Social Influence. By creating a professional network for verified farmers, it digitizes the trusted peer-to-peer knowledge sharing identified in the survey and provides a safe, productive alternative to open social media platforms.